SCHOOL BUS TRACKING SYSTEM

Detailed Micro-Phase Development Roadmap

55 major phases • 211 micro-phases • 821+ implementation tasks

Purpose

This document converts the agreed school-bus tracking product into a buildable sequence of very small, testable phases. The goal is to build the core product first, validate it with real school-bus operations, and add commercial/payment functionality only at the end.

Development philosophy

- ☐ Free/open-source tools first wherever they are good enough.
- ☐ Use APIs and managed services when they save time or improve reliability.
- ☐ Keep the core business logic and database under our control.
- ☐ Use provider abstractions so maps, routing, notifications and storage can be changed later.
- ☐ Do not build unnecessary infrastructure from scratch.
- ☐ Do not add payments/subscriptions/invoices until the core transport product is stable.

Current scope rule

Payments, subscriptions, invoices, renewal and non-payment restrictions are deliberately postponed to the final commercial phase.

1. Final Product Architecture

The system is centered around one backend and one database. Driver GPS can initially come from the Android phone and later from certified/compatible GPS hardware without changing the rest of the product.

User / Source	Interface	Core system
Driver	Flutter Android + WhatsApp	Trip + GPS + Attendance
Parent	WhatsApp + secure live web page	ETA + notifications + child journey
School	Next.js/React dashboard	Fleet + routes + attendance + alerts + reports
Future GPS hardware	Device protocol	Same GPS ingestion API
Core backend	Node.js + TypeScript	Trip / GPS / ETA / Pickup / Delay / Notification
Data	PostgreSQL + PostGIS	Operational + geospatial data
Optional later	Redis / queues	Caching + real-time state + background work

2. How to Use This Roadmap

☐ Work sequentially unless a dependency is explicitly independent.
☐ A micro-phase is complete only when its tasks work and the relevant test passes.
☐ Do not jump into advanced intelligence before GPS, trips and data quality are stable.
☐ Keep a single master Feature Register updated after every completed micro-phase.
☐ For every production feature, define: database change → API → UI/WhatsApp flow → error states → test.
☐ Keep payment/commercial work parked until the end.

PHASE 0 — Project Control & Scope Lock

0.1 — Master scope register

☐ Create one master feature list

☐ Assign unique Feature IDs

☐ Mark each feature as Core / Safety / Supporting

☐ Record dependencies

☐ Create status values: Not Started / In Progress / Blocked / Done

0.2 — Scope exclusions

☐ Keep payments, subscriptions, invoices and billing out of current build

☐ Keep CCTV, RFID, face recognition, fuel, payroll and ERP out of current MVP

☐ Keep custom GPS hardware out of first software build

☐ Keep iOS driver app out of initial build

0.3 — Product rules

☐ WhatsApp-first for simple parent/driver actions

☐ Parent does not need a mandatory app

☐ Driver app remains minimal and handles GPS/background location

☐ Free/open-source first; paid only when necessary

0.4 — Definition of done

☐ Every feature must have UI/API/database behavior defined

☐ Every critical feature must have a test case

☐ No feature is marked Done without a real end-to-end test

PHASE 1 — Development Environment

1.1 — Developer workstation

☐ Install Git

☐ Install VS Code

☐ Install Node.js

☐ Install Flutter SDK

☐ Install Android Studio

☐ Install Docker

1.2 — Repository

☐ Create GitHub repository

☐ Create main branch and development branch

☐ Create README

☐ Add issue templates

☐ Add environment variable template

1.3 — Monorepo structure

☐ Create backend folder

☐ Create driver-app folder

☐ Create school-dashboard folder

☐ Create parent-web folder

☐ Create shared folder

☐ Create infrastructure folder

☐ Create docs folder

☐ Create tests folder

1.4 — Local services

☐ Run PostgreSQL locally

☐ Enable PostGIS

☐ Run Redis locally when needed
☐ Create Docker Compose for local dependencies

PHASE 2 — Database & Data Model

2.1 — Core entities

☐ Schools
☐ Admins
☐ Drivers
☐ Buses
☐ Students
☐ Parents

2.2 — Transport entities

☐ Routes
☐ Pickup locations
☐ Trips
☐ GPS locations
☐ Boarding events
☐ Drop events

2.3 — Operations entities

☐ Notifications
☐ Delays
☐ Incidents
☐ SOS events
☐ Temporary pickup requests
☐ Absence records
☐ Reports
☐ Audit logs

2.4 — Location model

☐ Store latitude/longitude
☐ Use PostGIS geometry
☐ Add spatial indexes
☐ Support distance and geofence queries

2.5 — Database quality

☐ Primary/foreign keys
☐ Unique constraints
☐ Indexes
☐ Created/updated timestamps
☐ Soft-delete strategy where useful
☐ Migration system
☐ Seed data

PHASE 3 — Backend Foundation

3.1 — API skeleton

☐ Create Node.js + TypeScript backend
☐ Create API router
☐ Create /health endpoint
☐ Create configuration loader

3.2 — Validation

☐ Request validation
☐ Response schemas
☐ Common error format
☐ Input sanitization

3.3 — Infrastructure services
☐ Database service
☐ Logging service
☐ Time/date utility
☐ ID generation
☐ File/storage abstraction

3.4 — API quality
☐ API versioning
☐ Rate-limit abstraction
☐ Request IDs
☐ Central error handler

PHASE 4 — Authentication & Authorization

4.1 — Admin authentication
☐ Login
☐ Logout
☐ Password hashing
☐ Forgot password
☐ Reset password
☐ Change password
☐ Session expiry

4.2 — Driver authentication
☐ Driver login
☐ Session handling
☐ Logout
☐ Assigned-bus authorization

4.3 — Parent identity
☐ WhatsApp identity mapping
☐ Secure parent token
☐ Secure tracking link
☐ Token expiry/revocation

4.4 — Roles & isolation
☐ School admin role
☐ Driver role
☐ Parent role
☐ School-to-school data isolation
☐ Driver restricted to assigned bus/students
☐ Parent restricted to authorized student data

4.5 — Security audit
☐ Audit important actions
☐ Protect secrets
☐ HTTPS in deployed environments
☐ Backup policy
☐ Basic abuse/rate-limit protections

PHASE 5 — School Onboarding & Admin

5.1 — School profile

- ☐ School name
- ☐ Address
- ☐ City
- ☐ State
- ☐ Contact
- ☐ Email
- ☐ Logo

5.2 — Admin profile

- ☐ Admin mobile
- ☐ Login credentials
- ☐ Profile
- ☐ Password change
- ☐ Logout

5.3 — Dashboard shell

- ☐ Sidebar
- ☐ Header
- ☐ Navigation
- ☐ Protected pages
- ☐ Responsive layout
- ☐ Empty states

5.4 — Admin home

- ☐ Dashboard cards
- ☐ Recent activity area
- ☐ Alert summary
- ☐ Transport status summary

PHASE 6 — Bus Management

6.1 — Create bus

- ☐ Bus number
- ☐ Registration number
- ☐ Bus details
- ☐ Save validation

6.2 — Assignments

- ☐ Assign driver
- ☐ Assign route
- ☐ Show assignments

6.3 — Bus details

- ☐ View bus
- ☐ Edit bus
- ☐ Remove bus
- ☐ Current location
- ☐ Current trip

6.4 — Bus history

- ☐ Trip history
- ☐ Incident history
- ☐ Attendance history

PHASE 7 — Driver Management

7.1 — Create driver
☐ Name
☐ Mobile
☐ Photo
☐ Login

7.2 — Assignment
☐ Assign bus
☐ Show assigned bus
☐ Show assigned route

7.3 — Driver history
☐ Trip history
☐ Boarding activity
☐ Incident history

7.4 — Driver profile
☐ View
☐ Edit
☐ Credential/session controls

PHASE 8 — Student Management

8.1 — Create student
☐ Name
☐ Class
☐ Section
☐ Student ID

8.2 — Parent/guardian data
☐ Guardian name
☐ Guardian phone
☐ Emergency contact

8.3 — Transport assignment
☐ Assigned bus
☐ Pickup location
☐ Transport status

8.4 — Student records
☐ View
☐ Edit
☐ Remove
☐ Transport history

PHASE 9 — Parent Management & Onboarding

9.1 — Parent record
☐ Parent name
☐ Mobile
☐ Relationship
☐ WhatsApp status

9.2 — Student assignment
☐ Link authorized student

☐ Show assignment

☐ Communication history

9.3 — WhatsApp onboarding

☐ Invitation flow

☐ Identity verification

☐ Activation confirmation

☐ Invalid/expired token handling

9.4 — Notification status

☐ Track notification delivery state

☐ Show communication history

PHASE 10 — Driver App Foundation

10.1 — Flutter app shell

☐ Login screen

☐ Home screen

☐ Today's bus

☐ Today's route

10.2 — Device permissions

☐ Location permission

☐ Background location permission

☐ Notification permission where required

10.3 — Driver status

☐ GPS status

☐ Network status

☐ Battery status

☐ Trip status

10.4 — Reliability

☐ Reconnect logic

☐ App lifecycle handling

☐ Last-known state

☐ Local pending queue

PHASE 11 — GPS Capture & Ingestion

11.1 — Location capture

☐ Latitude

☐ Longitude

☐ Timestamp

☐ Speed

☐ Accuracy

☐ Battery

☐ Network

11.2 — Upload

☐ Send location to backend

☐ Validate payload

☐ Retry failed upload

☐ Avoid duplicate points

11.3 — GPS health

☐ Detect stale GPS

☐ Detect poor accuracy

☐ Detect internet loss

☐ Store last known location

11.4 — Backend processing

☐ Validate source

☐ Store raw location

☐ Update current bus position

☐ Prepare real-time feed

PHASE 12 — Real-Time Live Tracking

12.1 — Current bus state
☐ Current location
☐ Last update time
☐ Speed
☐ Trip state

12.2 — Real-time transport
☐ WebSocket/SSE or equivalent
☐ Reconnect
☐ Subscription by school/bus

12.3 — School map feed
☐ Bus marker
☐ Driver
☐ Route
☐ Trip status
☐ Last update

12.4 — Parent live feed
☐ Authorized bus only
☐ Secure live location access
☐ Graceful offline state

PHASE 13 — Map & Routing Layer

13.1 — Map abstraction
☐ Create MapService interface
☐ Create RoutingService interface
☐ Keep provider-specific code isolated

13.2 — Free-first maps
☐ OpenStreetMap data
☐ Map rendering option
☐ Geocoding option
☐ Routing option

13.3 — Routing
☐ OSRM/Valhalla evaluation
☐ Route distance
☐ Road path
☐ Fallback behavior

13.4 — Provider fallback
☐ Document when paid maps may be used
☐ Keep provider switch possible
☐ Track map API usage

PHASE 14 — Dynamic Pickup Engine

14.1 — Parent location sharing
☐ WhatsApp asks for location
☐ Receive coordinates
☐ Validate coordinates

14.2 — Pickup point creation
☐ Find nearest suitable road
☐ Reverse geocode

☐ Create readable address

☐ Store coordinates

14.3 — Student mapping

☐ Assign pickup to student

☐ Show pickup on map

☐ Update pickup

14.4 — Safety/quality checks

☐ Reject clearly invalid locations

☐ Detect duplicate/nearby pickup points

☐ Allow school to edit/change pickup

PHASE 15 — Route Management

15.1 — Create route

☐ Route name

☐ Assign bus

☐ Assign driver

15.2 — Pickup sequence

☐ Add pickup locations

☐ Order pickup sequence

☐ Edit sequence

☐ Remove pickup

15.3 — Route map

☐ Draw route

☐ Show pickup points

☐ Show route details

15.4 — Route history

☐ Track changes

☐ Show previous route state where required

PHASE 16 — Trip Engine

16.1 — Trip lifecycle

☐ NOT_STARTED

☐ STARTED

☐ IN_PROGRESS

☐ ARRIVED_SCHOOL

☐ COMPLETED

16.2 — Trip creation

☐ Create today's trip

☐ Attach bus

☐ Attach driver

☐ Attach route

16.3 — Trip start

☐ Start timestamp

☐ Activate tracking session

☐ Update dashboard

16.4 — Trip end

☐ End timestamp

☐ Stop trip state

☐ Lock trip attendance
☐ Generate summary

16.5 — Idempotency
☐ Prevent duplicate starts
☐ Prevent duplicate ends
☐ Handle retries safely

PHASE 17 — Driver WhatsApp Core

17.1 — Menu
☐ Start Trip
☐ End Trip
☐ Student Search
☐ Report Problem
☐ Emergency
☐ Help

17.2 — Start Trip flow
☐ Identify driver
☐ Identify assigned bus
☐ Create/start trip
☐ Activate GPS session
☐ Trigger parent notifications
☐ Update school dashboard

17.3 — End Trip flow
☐ Identify active trip
☐ End trip
☐ Stop session
☐ Lock attendance
☐ Generate trip summary

17.4 — Error handling
☐ No assigned bus
☐ No active trip
☐ Duplicate command
☐ Unknown command
☐ Temporary WhatsApp/API failure

PHASE 18 — Driver Attendance

18.1 — Student list
☐ Show registered students for assigned bus
☐ Show pickup details

18.2 — Boarding
☐ Boarded
☐ Not boarded
☐ Timestamp
☐ Location

18.3 — Drop
☐ Dropped
☐ Timestamp
☐ Location

18.4 — History
☐ Trip/date history
☐ Student journey history
☐ Correction rules

PHASE 19 — Driver Student Search

19.1 — Search
☐ Search by student name
☐ Restrict results to assigned bus

19.2 — Student result
☐ Student name
☐ Parent name
☐ Parent phone
☐ Pickup address
☐ Pickup location
☐ Bus
☐ Route

19.3 — Actions
☐ Call parent
☐ Open map/navigation
☐ Return to menu

PHASE 20 — ETA Engine

20.1 — ETA inputs
☐ Current bus position
☐ Route
☐ Pickup locations
☐ Road distance
☐ Current/estimated speed

20.2 — ETA calculation
☐ Bus-to-pickup ETA
☐ Next pickup ETA
☐ Remaining route ETA

20.3 — ETA refresh
☐ Refresh after GPS updates
☐ Smooth noisy estimates
☐ Handle stopped bus

20.4 — Parent ETA
☐ Generate parent-specific ETA
☐ Show last update
☐ Handle unavailable ETA

PHASE 21 — Geofencing

21.1 — Pickup geofence
☐ Create radius/shape
☐ Detect entry
☐ Detect exit

21.2 — School geofence
☐ Create school geofence
☐ Detect school arrival

21.3 — Event handling
☐ Debounce repeated events
☐ Store timestamp/location
☐ Trigger workflows

PHASE 22 — Parent WhatsApp

22.1 — Parent menu
☐ Track My Bus
☐ My ETA
☐ My Child
☐ My Pickup
☐ Notifications
☐ Help

22.2 — Track My Bus
☐ Current bus
☐ Location
☐ Route
☐ Pickup
☐ ETA

☐ Trip status
☐ Secure live link

22.3 — My ETA
☐ Return latest ETA
☐ Return last update
☐ Handle unavailable ETA

22.4 — My Child
☐ Current journey status
☐ Boarding status
☐ Drop status

22.5 — My Pickup
☐ View pickup
☐ Request pickup update flow

22.6 — Help
☐ FAQ
☐ Support contact
☐ Issue reporting

PHASE 23 — Parent Live Web Page

23.1 — Secure entry
☐ Token validation
☐ Expiry
☐ Revocation

23.2 — Live view
☐ Bus marker
☐ Route
☐ Pickup
☐ ETA
☐ Bus number
☐ Trip status
☐ Last update

23.3 — Failure states
☐ Bus offline
☐ Trip ended
☐ Invalid link
☐ Stale location

PHASE 24 — Notification Engine

24.1 — Event model
☐ Define notification events
☐ Define recipients
☐ Define priority

24.2 — Parent events
☐ Bus started
☐ Approaching
☐ 10-minute alert
☐ 3-minute/nearby alert
☐ Reached pickup

☐ Child boarded
☐ Child dropped
☐ Trip completed

24.3 — Operational events
☐ Delay
☐ GPS unavailable
☐ Route deviation
☐ Emergency
☐ Unusual stop

24.4 — Delivery
☐ WhatsApp
☐ Push where appropriate
☐ Dashboard alerts
☐ Retry/failure state

24.5 — Anti-spam
☐ Deduplicate events
☐ Throttle repeated alerts
☐ Only notify affected parents

PHASE 25 — Smart Delay Engine

25.1 — Expected timeline
☐ Calculate expected ETA for each pickup
☐ Store expected arrival

25.2 — Actual timeline
☐ Compare current ETA
☐ Calculate delay minutes

25.3 — Affected parent logic
☐ Identify upcoming affected pickups
☐ Exclude already-served pickups
☐ Identify parents still waiting

25.4 — Delay notifications
☐ Send delay alert
☐ Update ETA
☐ Send recovery update when appropriate

25.5 — Spam control
☐ Avoid repeated identical delay messages
☐ Notify only on meaningful change

PHASE 26 — Long Stop Detection

26.1 — Detection
☐ Detect stationary bus
☐ Configurable threshold
☐ Ignore GPS jitter

26.2 — Driver reason
☐ Traffic
☐ Vehicle problem
☐ Student issue
☐ Road block

☐ Other

☐ No problem

26.3 — School workflow

☐ Create event

☐ Notify school

☐ Store reason

☐ Resolve event

PHASE 27 — GPS Offline & Recovery

27.1 — Detection

☐ Stale location threshold

☐ Internet loss

☐ App disconnect

27.2 — Response

☐ Store last known location

☐ Notify school

☐ Notify driver

27.3 — Recovery

☐ Reconnect automatically

☐ Upload queued points where valid

☐ Mark recovered

☐ Avoid duplicate points

PHASE 28 — Route Deviation

28.1 — Expected route

☐ Load route geometry

☐ Determine bus position relative to route

28.2 — Detection

☐ Distance threshold

☐ Debounce

☐ Create deviation event

28.3 — Response

☐ School alert

☐ Incident record

☐ Current location

☐ Timestamp

PHASE 29 — Overspeed

29.1 — Rule

☐ Configurable speed threshold

☐ Optional duration threshold

29.2 — Detection

☐ Use validated GPS speed

☐ Avoid single-point false positives

29.3 — Event

☐ Speed

☐ Time

☐ Location

☐ Duration

29.4 — Reporting

☐ School notification

☐ Daily report inclusion

☐ History

PHASE 30 — Emergency / SOS

30.1 — Trigger
☐ Driver WhatsApp emergency
☐ Optional app emergency control

30.2 — Capture
☐ Bus
☐ Driver
☐ GPS
☐ Timestamp
☐ Active trip

30.3 — School alert
☐ Immediate alert
☐ Live bus location
☐ Driver details
☐ Incident record

30.4 — Resolution
☐ Acknowledge
☐ Resolve
☐ Store incident history

30.5 — Scope note
☐ Keep software SOS separate from physical AIS-140 panic hardware

PHASE 31 — Pickup Arrival Workflow

31.1 — Upcoming student
☐ Determine next pickup
☐ Identify associated student

31.2 — Approach
☐ Approaching notification
☐ Driver upcoming-student view

31.3 — Arrival
☐ Geofence event
☐ Parent arrival notification
☐ Boarding available

PHASE 32 — Student Journey Record

32.1 — Journey timeline
☐ Pickup
☐ Bus arrival
☐ Boarded/not boarded
☐ On route
☐ School arrival
☐ Dropped

32.2 — History
☐ Student transport history
☐ Trip history
☐ Attendance history

32.3 — Data consistency

☐ One journey per trip/student
☐ Prevent duplicate attendance events

PHASE 33 — Student Not Going Today

33.1 — Parent flow
☐ Select student
☐ Select date
☐ Confirm absence

33.2 — System response
☐ Mark transport absence
☐ Notify school/driver
☐ Update driver's student list

33.3 — Trip behavior
☐ Do not expect boarding alert for absent student
☐ Keep audit record

PHASE 34 — Temporary Pickup Change

34.1 — Request
☐ Parent chooses date
☐ Parent shares new location
☐ Create request

34.2 — School review
☐ View request
☐ Accept
☐ Reject
☐ Record decision

34.3 — Driver update
☐ Update route/pickup information
☐ Show temporary pickup clearly

34.4 — Parent confirmation
☐ Notify parent of final decision
☐ Show effective date

PHASE 35 — School Live Dashboard

35.1 — Summary
☐ Running buses
☐ Completed trips
☐ Delayed
☐ Offline
☐ Incidents
☐ Boarding/drop overview

35.2 — Bus list
☐ Bus status
☐ Driver
☐ Route
☐ Trip state
☐ Last GPS update

35.3 — Quick actions
☐ Open live map
☐ Open bus details
☐ Open alerts

PHASE 36 — School Live Map

36.1 — Fleet map
☐ All authorized buses
☐ Markers
☐ Current location

36.2 — Bus details
☐ Driver
☐ Trip
☐ Student count/attendance overview
☐ ETA

36.3 — Map filters
☐ Running
☐ Delayed
☐ Offline
☐ Incidents

PHASE 37 — Attendance Dashboard

37.1 — Boarding view
☐ Boarded
☐ Not boarded
☐ Timestamp
☐ Location

37.2 — Drop view
☐ Dropped
☐ Timestamp
☐ Location

37.3 — Filters
☐ Date
☐ Bus
☐ Route
☐ Driver
☐ Student

37.4 — History
☐ Trip history
☐ Student history

PHASE 38 — Daily Transport Reports

38.1 — Report inputs
☐ School
☐ Date
☐ Bus
☐ Driver

☐ Route

38.2 — Trip metrics

☐ Trip start
☐ School arrival
☐ Expected
☐ Boarded
☐ Not boarded
☐ Dropped

38.3 — Safety/quality metrics

☐ Delay
☐ Route deviation
☐ Overspeed
☐ GPS downtime
☐ Incidents
☐ Emergency

38.4 — Report generation

☐ Dashboard report
☐ CSV export

PHASE 39 — Report & Summary Delivery

39.1 — Daily summary

☐ Generate daily summary
☐ Send to authorized school recipients

39.2 — Weekly summary

☐ Aggregate trips
☐ Delays
☐ GPS health
☐ Attendance
☐ Incidents

39.3 — Delivery tracking

☐ Generated
☐ Sent
☐ Failed
☐ Retry

PHASE 40 — School Alert Center

40.1 — Alert types
☐ Emergency
☐ Route deviation
☐ Overspeed
☐ GPS offline
☐ Long stop
☐ Delay

40.2 — Alert lifecycle
☐ New
☐ Acknowledged
☐ Resolved

40.3 — Incident linkage
☐ Create incident from alert
☐ Attach evidence/data
☐ Store resolution

PHASE 41 — Support System

41.1 — Help
☐ FAQ
☐ How-to guides
☐ Contact support

41.2 — Issue reporting
☐ Technical issue
☐ Category
☐ Description
☐ Related bus/trip

41.3 — Tickets
☐ Create
☐ Track
☐ Resolve
☐ History

PHASE 42 — Internal Operations Panel

42.1 — Customer operations
☐ School list
☐ School details
☐ Bus count
☐ Customer status

42.2 — System health
☐ GPS health
☐ WhatsApp delivery health
☐ API health
☐ Error logs

42.3 — Support
☐ Support tickets
☐ Activity logs

42.4 — Access control
☐ Internal admin roles
☐ Audit sensitive actions

PHASE 43 — Monitoring & Reliability

43.1 — Application monitoring
☐ Logs
☐ Error tracking
☐ Health checks

43.2 — Infrastructure
☐ Database health
☐ CPU/memory where available
☐ Storage
☐ Backups

43.3 — Product monitoring
☐ GPS uptime
☐ WhatsApp failures
☐ Notification failures
☐ ETA failures

43.4 — Alerts
☐ Critical error alerts
☐ Database alerts
☐ Service downtime alerts

PHASE 44 — Analytics

44.1 — Parent usage
☐ Track requests
☐ ETA requests
☐ Weekly usage

44.2 — Driver usage
☐ Start Trip success
☐ End Trip success
☐ Student search
☐ Attendance completion

44.3 — System metrics
☐ GPS uptime
☐ ETA accuracy
☐ Notification delivery
☐ Delay detection accuracy

44.4 — Privacy-aware analytics
☐ Avoid unnecessary personal data
☐ Define retention
☐ Use aggregated metrics where possible

PHASE 45 — Testing — Unit & Integration

45.1 — Backend unit tests
☐ Trip state

☐ ETA calculations
☐ Delay calculations
☐ Geofence logic
☐ Permissions

45.2 — API integration tests

☐ Authentication
☐ Bus APIs
☐ Student APIs
☐ GPS APIs
☐ Trip APIs
☐ Notification APIs

45.3 — Database tests

☐ Constraints
☐ Spatial queries
☐ Migrations
☐ Seed data

PHASE 46 — Testing — End-to-End

46.1 — Driver flow

☐ Login
☐ Start Trip
☐ GPS
☐ Search student
☐ Boarding
☐ Problem
☐ SOS
☐ End Trip

46.2 — Parent flow

☐ Onboarding
☐ Track bus
☐ ETA
☐ Pickup
☐ Child status
☐ Delay
☐ Absence
☐ Temporary pickup

46.3 — School flow

☐ Login
☐ Manage bus/driver/student
☐ Route
☐ Live map
☐ Attendance
☐ Alerts
☐ Reports

PHASE 47 — Failure & Recovery Testing

47.1 — Driver device failures

☐ GPS off

☐ Internet off
☐ Low battery
☐ App killed
☐ Phone restart

47.2 — Backend failures
☐ Backend restart
☐ Database unavailable
☐ Queue/retry behavior

47.3 — Integration failures
☐ WhatsApp delayed
☐ Notification failure
☐ Routing failure

47.4 — Data safety
☐ Duplicate GPS
☐ Duplicate trip start
☐ Duplicate trip end
☐ Invalid/stale location

PHASE 48 — Security Testing

48.1 — Authorization
☐ School A cannot access School B
☐ Driver cannot access other bus
☐ Parent cannot access unauthorized student

48.2 — Application security
☐ Input validation
☐ Rate limits
☐ Secure sessions
☐ Secret handling

48.3 — Privacy
☐ Minimum necessary personal data
☐ Secure tracking links
☐ Retention rules
☐ Audit access

48.4 — Backup recovery
☐ Backup creation
☐ Restore test
☐ Document recovery steps

PHASE 49 — Pilot Preparation

49.1 — Pilot configuration
☐ Select 1 school
☐ Configure 1–3 buses
☐ Register real drivers
☐ Register sample/real parent users with permission

49.2 — Operational readiness
☐ Driver training
☐ School admin training
☐ Parent onboarding instructions

☐ Support contact

49.3 — Pilot dashboards

☐ GPS uptime
☐ ETA accuracy
☐ Trip-start success
☐ WhatsApp delivery
☐ Attendance completion
☐ Parent usage

PHASE 50 — Pilot & Improvement Loop

50.1 — Live pilot
☐ Run real trips
☐ Collect issues
☐ Monitor reliability

50.2 — Bug triage
☐ P0 safety/system outage
☐ P1 major feature
☐ P2 normal bug
☐ P3 minor UI

50.3 — Iteration
☐ Fix
☐ Regression test
☐ Deploy
☐ Verify with real users

50.4 — Acceptance
☐ GPS target >95% usable uptime
☐ Initial ETA target roughly ±5–10 minutes
☐ High trip-start reliability
☐ Usable driver workflow
☐ Useful parent alerts

PHASE 51 — Production Hardening

51.1 — Performance
☐ Database indexes
☐ Query optimization
☐ Caching
☐ Pagination

51.2 — Reliability
☐ Retries
☐ Idempotency
☐ Queues where needed
☐ Graceful degradation

51.3 — Operations
☐ Automated backups
☐ Restore drills
☐ Monitoring
☐ Incident runbook

51.4 — Deployment
☐ Staging-to-production process
☐ Versioning
☐ Rollback plan
☐ Environment separation

PHASE 52 — GPS Hardware Integration — Later

52.1 — Hardware abstraction
☐ Define common GPS ingestion contract

☐ Accept phone GPS source
☐ Prepare hardware GPS source

52.2 — AIS-140/VLTD evaluation
☐ Select compliant hardware options
☐ Evaluate data protocol
☐ Evaluate SIM/network
☐ Evaluate backup power

52.3 — Integration
☐ Hardware → same GPS API
☐ Device identification
☐ Health monitoring
☐ Fallback/last-known behavior

PHASE 53 — Scale & Cost Optimization — Later

53.1 — Routing scale
☐ Evaluate self-hosted OSRM/Valhalla
☐ Measure routing cost
☐ Move only when justified

53.2 — Infrastructure scale
☐ Dedicated database if needed
☐ Queues
☐ Redis
☐ Load handling
☐ Backups

53.3 — Provider abstraction
☐ Maps provider switch
☐ Notification provider switch
☐ Storage provider switch

53.4 — Cost monitoring
☐ Track cloud usage
☐ Track API usage
☐ Track message usage
☐ Identify expensive operations

PHASE 54 — Payments & Commercial Layer — LAST

54.1 — Subscription model
☐ School subscription
☐ Bus count pricing
☐ Monthly billing

54.2 — Payments
☐ Payment gateway integration
☐ Payment status
☐ Webhook handling

54.3 — Invoices
☐ Invoice generation
☐ Payment history
☐ Renewal

54.4 — Account controls

☐ Expiry handling

☐ Grace period

☐ Restrictions after non-payment

54.5 — Commercial analytics

☐ MRR

☐ School count

☐ Bus count

☐ Churn

☐ Payment failures

Final Scope Checklist

The following agreed product areas are all represented in the roadmap:

- ☐ School onboarding and admin dashboard
- ☐ Bus, driver, student and parent management
- ☐ Dynamic parent pickup location
- ☐ Route creation and pickup sequencing
- ☐ Minimal driver Android app with background GPS
- ☐ Driver WhatsApp: Start Trip, End Trip, Student Search, Report Problem, Emergency, Help
- ☐ Driver boarding/drop attendance
- ☐ Live GPS and real-time fleet tracking
- ☐ Parent WhatsApp: Track My Bus, My ETA, My Child, My Pickup, Notifications, Help
- ☐ Parent secure live tracking page
- ☐ ETA engine
- ☐ Pickup and school geofencing
- ☐ Bus approaching / pickup arrival notifications
- ☐ Boarding and drop notifications
- ☐ Smart delay detection and affected-parent notifications
- ☐ Long-stop detection and reason capture
- ☐ GPS offline detection and recovery
- ☐ Route deviation detection
- ☐ Overspeed detection
- ☐ Emergency/SOS workflow
- ☐ Student journey history
- ☐ Student not going today / absence
- ☐ Temporary pickup change
- ☐ School live dashboard and map
- ☐ Attendance dashboard
- ☐ Daily and weekly transport reports
- ☐ School alert center
- ☐ Support system
- ☐ Internal operations panel
- ☐ Monitoring and analytics
- ☐ Unit, integration, end-to-end, failure and security testing
- ☐ Pilot deployment and improvement loop
- ☐ Production hardening
- ☐ Later GPS hardware integration
- ☐ Later scale/cost optimization
- ☐ Payments and commercial layer — LAST

Explicitly excluded from the current scope

- ☐ Bus capacity
- ☐ Bus active/inactive field
- ☐ Parent child-selection feature requested for removal
- ☐ One parent WhatsApp number → multiple children workflow
- ☐ General mandatory school-admin approval for normal pickup setup
- ☐ CCTV / AI camera
- ☐ RFID / face recognition
- ☐ Fuel / payroll / full ERP
- ☐ Custom GPS hardware manufacturing
- ☐ iOS driver app in the initial build

☐ Payments/subscriptions/invoices until the final commercial phase

Recommended build order

Foundation → database → authentication → school data → driver app → GPS → live tracking → routes/pickups → trip engine → driver WhatsApp → attendance → ETA → parent WhatsApp → notifications → delay/safety intelligence → school dashboard/reports → testing → pilot → hardening → hardware/scale → payments.

Important: this roadmap is the working master plan. If a new feature is added later, assign it a new Feature ID and insert it into the appropriate phase rather than silently changing the scope.